

Rungs, Voids and Effective Closure:

Cognitive Spaces, Uncertified Convergence and Promotion

Nial Goodman

nialgoodman@gmail.com

Formalisation developed with Claude, an AI system by Anthropic (see Author contributions)

Abstract. We separate problems that are merely difficult for an agent from problems in its *void*, beyond anything it can approach. A cognitive space is a Turing ideal; *effective closure* adds the limits an agent can approach by approximations it can generate. Without effectivity closure is trivial: boundaries are properties of agents, not of the topology. Iterating effective closure from the computable sets gives exactly the finite levels of the arithmetical hierarchy, and the ladder exists only because convergence is uncertified: requiring a modulus makes closure idempotent. Bounded mind changes and probabilistic convergence add no rungs, and voids are generic: the closure of a countable space is meagre and null. The union of the finite rungs is a fixed point excluding true arithmetic, so convergence stalls at ω and crossing requires a leap, recognisable neither from a process's rules (undecidable) nor from its behaviour: no observer of any power can recognise, even in the limit, that a convergent process has left the closure of a countable space, while next-rung content is recognisable in the limit from two rungs above. An agent has a certified and a committed space and climbs only by *promotion*, an uncertified commitment to its own limits, which changes nothing about what it computes and everything about which sets its beliefs converge to. Content crosses between agents only as method, as testimony, or as certification from a strictly higher rung. Encounters (merger, elimination, trade) are operations on a set of spaces; merger is the one certified route to new content inside a rung (by Sacks splitting, two agents who cannot settle the halting problem settle it by merging); and a belief that a peer lies within one's space can converge correctly, while no belief that a peer exceeds it can: subsumption is limit-verifiable, strict superiority is not.

Series note. This is Paper B, the core, of the series *Rungs, Voids and Effective Closure*, the first programme of the study *Effective Horizons and Cognitive Spaces* (synopsis in [1]), four working papers on a model of agents whose cognitive reach is a Turing ideal and who extend it by uncertified convergence. This paper supplies the framework: effective closure, the rung ladder, the stall at ω , the Indistinguishability Theorem, the agent model and the encounter application, together with the programme synthesis of Section 13. The companion paper [4] turns closure steps into a distance and studies its geometry (it can be skipped by readers without a background in logic); the sequels [2] and [3] add time, populations, finite resources and strategy, and take the resulting dynamics to the Fermi problem; Section 12 lists where each attaches to the core.

1 INTRODUCTION

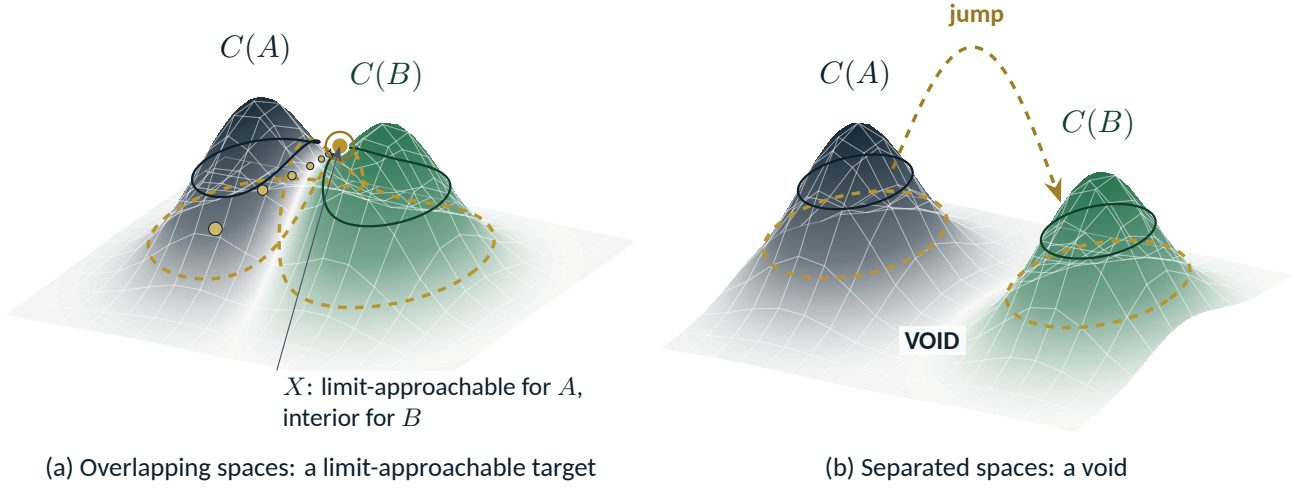


Figure 1: Two cognitive spaces, drawn as landscapes (the curvature is illustrative; the underlying space is Cantor space and has no such geometry). Solid contours bound the interiors, where targets are settled at finite cost; dashed gold contours mark the effective closures, and the band between is the boundary, approachable by convergence but never attained. (a) Where the spaces overlap, a target on A 's boundary is *limit-approachable* for A : a sequence of reachable approximations (gold points) converges to it. The same target lies in B 's interior. (b) Targets outside both closures lie in the pale floor between them, a *void* for both agents. No convergent path reaches such a target, and passage requires a jump. (Two closures are never disjoint, since each contains the computable sets; the picture shows the region neither reaches.)

Fermi's question, why an apparently hospitable galaxy is silent, has attracted answers that depend almost entirely on unmeasured priors [16, 31]. One of the more influential in popular culture is the "dark forest" of Liu [21]: civilisations that reveal themselves are destroyed, so survivors hide. Liu's argument is game-theoretic, and it rests on an assumption it never states: that civilisations share enough cognitive ground to recognise one another as players, model one another's options, and reason about strikes.

This paper asks when that assumption fails, and answers in one well-understood setting. We introduce the notion of a *cognitive space*, the region of problem space in which an agent has usable cognitive function, and ask how two such spaces can be separated. Several kinds of separation must be distinguished. Some targets are *costly*: settled in principle, at a cost that may be astronomical; the four-colour theorem is verified but not surveyable by any human. Others are *limit-approachable*: the agent can generate approximations that converge to them but can never attain or certify them. Still others lie in a *void*: no expenditure of the agent's resources approaches them at all (Figure 1). We use "hard" informally for the first two together, and distinguish them whenever it matters. A squirrel acts in accordance with Euclidean geometry when it judges a leap, but it will never represent the Pythagorean theorem.

The squirrel marks the limit of what this paper covers, not an instance of it. The model developed here begins at Turing universality. It therefore does not distinguish two oracle-free Turing-universal architectures by their finite speed, memory, embodiment or ease of discovering algorithms: under Postulate 1 they occupy the same certified space R_0 . The distinctions studied here are between agents with different oracle resources, agents at different depths of uncertified commitment (Section 6), and lineages compared under repeated effective closure. Architectural limitations below universality, the intuitive source of examples such as a squirrel that cannot formulate algebra, require a resource-bounded or subrecursive analogue of the construction and remain an open extension (Section 14, item 1). Such examples motivate that extension; they do not instantiate the present theorems.

The central difficulty is to make "no expenditure approaches them" precise without making it trivial. Our answer is that the relevant notion of approach is *effective and uncertified convergence*: an agent approaches a target when it can itself generate approximations that eventually stabilise on the target, without being able to certify when they have done so. Computability theory already contains the machinery to analyse this notion, and the analysis produces a ladder of rungs, an exact criterion separating limit-approachable targets from voids, and a proof that convergence alone stalls.

Summary of results.

- A set of postulates (Section 2) that fix what we take a cognitive space, a boundary and a void to be.
- An effective closure operator and four results (Section 4): a triviality lemma, the Rung Theorem, a uniqueness proposition locating the origin of rungs in uncertified convergence, and the Stall Theorem; together with the genericity of voids and two collapse results showing that neither bounded mind changes nor probabilistic convergence produce rungs of their own.
- A formal notion of *jump*, a proof that jumping is undecidable, and the *Indistinguishability Theorem*: no observer of any power can recognise a void even in the limit, while next-rung content is recognisable in the limit from two rungs above (Section 5).
- An agent model in which an agent carries a certified and a committed space, which separates settling, limit-identification and holding (belief at a commitment depth), identifies *promotion* as the uncertified step by which agents climb (an agent of depth k is a computable process whose beliefs are k -fold limits), shows that content crosses between agents only as algorithm, as method, as testimony or as certification from a strictly higher rung, that teaching transmits competence through marking rather than example, and places competence, mastery, certification and examination at successive rungs f , f or f' , f' and f'' (Section 6).
- A summary of the geometry of the rung distance on arithmetical agents, proved in the companion paper [4], and what it says about mutual intelligibility: within a component every separation is a finite lag, and permanent voids exist only between components (Section 7).
- One worked application (Section 8): merger, elimination and trade as operations on a set of spaces; merger as the one certified route to new content inside a rung (Sacks splitting), lateral and forgetful, with irreducible agents and non-principal common ground; trade priced at $d(x, y)$ uncertified promotions against a merger cost of zero; and the unverifiability of strict superiority, an asymmetry that says exactly what the dark-forest players cannot know.
- A transfinite summary (Section 9): the ladder continues along Kleene's $\mathcal{O}$ through the hyperarithmetical sets and meets its effective ceiling at ω_1^{CK} , the same horizon as Feferman's progressions.
- An explicit list of branching points (Section 12) at which the geometry [4] and the population and strategic analysis [2] attach to the core, and a programmatic synthesis of the series around three orderings, reachability, geometry and certification (Section 13).
- An interpretation (Section 10) reading the results as claims about inquiry, intelligibility and the Fermi problem, with one physical illustration of the certificate/limit distinction.
- An explicit list of open issues (Section 14).

2 FRAMEWORK AND POSTULATES

The theorems in Section 4 are standard; whether they concern minds depends entirely on the postulates below. Each can be challenged independently, and Section 14 records the principal objections.

Postulate 1 (Reachability). An agent's cognitive space is the class of targets it can settle given unbounded time and external memory, with its architecture held fixed. Formally, it is a Turing ideal (Definition 1).

This separates *content* gaps from *capacity* gaps. A medieval scholar lacked quantum mechanics as content; given instruction, it was reachable. The squirrel lacks algebra as capacity; that gap is sub-universal and lies outside the present model, as the introduction says.

Postulate 2 (Targets as decision problems). Targets are modelled as decision problems, that is, as subsets of $\mathbb{N}$.

This is the strongest simplification in the paper. It retains the functional content of cognition and discards the perspectival content (the difference between knowing *that* and knowing *what it is like*).

Postulate 3 (Effective approach). An agent approaches a target only through approximations it can itself generate.

Postulate 4 (Three regions). Relative to an agent, every target lies in its *interior* (settled at finite cost, which may be feasible or prohibitive), on its *boundary* (limit-approachable: approached by convergence but never attained or certified) or in its *void* (not approachable).

Postulate 5 (Rungs). A rung is a cognitive space closed under a fixed mode of approach. Rungs are ordered by inclusion: a higher rung embeds a lower one, while the lower sees the higher only through a lossy projection.

Postulate 6 (Traversal). Cognitive change is a sequence of cognitive spaces indexed by time. A step is continuous if its new content lies within the closure of the old, and a jump otherwise.

Postulates 1 and 3 encode a physical constraint we call *finite readout*. A physical agent measures to finite precision, in finite time, and must act on what it extracts. Whatever non-computable structure the world contains, an agent can exploit only what its channel delivers.

Remark 1 (The model is extensional). “ X lies in the agent’s space” means $X \leq_T A$ for the agent’s oracle A : a reduction exists. It does not mean the agent holds one, and no general method-discovery procedure is assumed. Where a target X is independently presented, as a stream against which outputs can be checked, candidate A -computable methods may be tested against that presentation and a correct one learned in the limit; without such a presentation the framework records the existence of a method and does not model its discovery. Every theorem of Sections 4–6 is extensional and unaffected. The idealisation bears on transmission as algorithm (Proposition 6) and on the interactive protocols of [2], where finding a witness for a reduction is shown to cost exactly one jump over possessing the reduction extensionally, and to be impossible at the agent’s own rung.

3 PRELIMINARIES

We use standard notation [33]. Identify $X \subseteq \mathbb{N}$ with its characteristic sequence in Cantor space $2^{\mathbb{N}}$, with metric

$$d(X, Y) = 2^{-\min\{n : X(n) \neq Y(n)\}}, \quad d(X, X) = 0.$$

Convergence $X_s \rightarrow X$ is pointwise stabilisation: for each n , $X_s(n) = X(n)$ for all sufficiently large s .

We write $X \leq_T A$ if $X = \Phi^A$ for some oracle machine Φ . The jump A' is the halting set relative to A ; by the relativised halting theorem, $A' \not\leq_T A$. Iterating from the empty set gives $\emptyset^{(0)} = \emptyset$ and $\emptyset^{(k+1)} = (\emptyset^{(k)})'$, and the ω -jump is the uniform join

$$\emptyset^{(\omega)} = \bigoplus_{k \in \mathbb{N}} \emptyset^{(k)} = \{\langle k, n \rangle : n \in \emptyset^{(k)}\}.$$

By Post’s theorem, $X \in \Delta_{k+1}^0$ if and only if $X \leq_T \emptyset^{(k)}$, and X is arithmetical if and only if $X \leq_T \emptyset^{(k)}$ for some k . The set $\emptyset^{(\omega)}$ is not arithmetical and is Turing-equivalent to true arithmetic [24, 35].

Definition 1 (Turing ideal). A *Turing ideal* is a non-empty class $R \subseteq 2^{\mathbb{N}}$ that is closed downward under $\leq_T$ and closed under $\oplus$. Cognitive spaces are taken to be Turing ideals throughout.

A sequence $(X_s)_{s \in \mathbb{N}}$ is *uniformly in* a class R if the function $g(n, s) = X_s(n)$, coded as a single set, belongs to R . For a Turing ideal this behaves as intended: the code of a constant sequence is Turing-equivalent to its value, and each row X_s is computable from g .

4 EFFECTIVE CLOSURE AND THE RUNG THEOREM

Definition 2 (Effective closure). For a Turing ideal R ,

$$\text{cl}(R) = \left\{ \lim_{s \rightarrow \infty} X_s : (X_s) \text{ is uniformly in } R \text{ and the limit exists} \right\}.$$

The *boundary* of R is $\partial R = \text{cl}(R) \setminus R$ and the *void* of R is $2^{\mathbb{N}} \setminus \text{cl}(R)$.

Constant sequences show $R \subseteq \text{cl}(R)$. This definition makes Postulate 4 exact.

Lemma 1. *If R is a Turing ideal then $\text{cl}(R) = \{X : X \leq_T A' \text{ for some } A \in R\}$, and $\text{cl}(R)$ is again a Turing ideal.*

Proof. If $g \in R$ generates X , then $X \leq_T g'$ by the limit lemma (Lemma 3 below). Conversely, if $X \leq_T A'$ with $A \in R$, the same lemma gives a generator $g \leq_T A$, and $g \in R$ by downward closure. The class $\{X : \exists A \in R, X \leq_T A'\}$ is closed downward, and closed under $\oplus$ because $A'_1 \oplus A'_2 \leq_T (A_1 \oplus A_2)'$ with $A_1 \oplus A_2 \in R$. ■

Lemma 2 (Effectivity is essential). *Without the uniformity requirement, every $X \subseteq \mathbb{N}$ is a limit of computable sets.*

Proof. Let $X_s = X \cap \{0, \dots, s-1\}$. Each X_s is finite, hence computable, and $X_s \rightarrow X$. ■

The computable sets are dense in $2^{\mathbb{N}}$. A boundary exists only because the agent must generate its own approximations: *boundaries are properties of agents, not of the topology.*

Corollary 1 (Voids are generic). *For every countable Turing ideal C (in particular every rung and A itself), $\text{cl}(C)$ is countable, dense, meagre and of Lebesgue measure zero. A target chosen by a fair coin lies in the void of C with probability one, and the voids of any countably many countable spaces have a comeagre, full-measure intersection.*

Proof. $\text{cl}(C)$ is a countable union of principal ideals by Lemma 1, each countable. Countable sets are meagre and null in Cantor space, and $\text{cl}(C) \supseteq R_0$ is dense. ■

Voids are therefore not the exception but the rule: what an agent can approach is a thin, dense skeleton in the space of targets, and the same is true of any countable family of agents taken together.

Lemma 3 (Limit lemma, relativised; Shoenfield [32]). *For every oracle A : $X \leq_T A'$ if and only if $X = \lim_s g(\cdot, s)$ for some $g \leq_T A$.*

Proof. ($\Leftarrow$) Fix n . The question “is there $t \geq s$ with $g(n, t) \neq g(n, s)$?” is Σ_1^0 relative to A and is therefore decided by A' . Query $s = 0, 1, 2, \dots$ until A' answers no, and output $g(n, s)$. The search terminates because $g(n, \cdot)$ stabilises.

($\Rightarrow$) Let $X = \Phi^{A'}$. Let A'_s be the set of $e < s$ such that $\Phi_e^A(e)$ halts within s steps; A'_s is computable from A uniformly in s , and $A'_s \rightarrow A'$. Let $g(n, s)$ be the output of $\Phi^{A'_s}(n)$ run for s steps, or 0 if it has not halted. The true computation $\Phi^{A'}(n)$ halts after some t steps with oracle use below some u . Once $s \geq t$ and A'_s agrees with A' below u , we have $g(n, s) = X(n)$. Thus $g \leq_T A$ and $g(\cdot, s) \rightarrow X$. ■

Theorem 1 (Rung Theorem). *Let R_0 be the class of computable sets and $R_{k+1} = \text{cl}(R_k)$. Then for every $k \in \mathbb{N}$,*

$$R_k = \{X : X \leq_T \emptyset^{(k)}\} = \Delta_{k+1}^0, \quad R_k \subsetneq R_{k+1}.$$

Proof. By induction on k ; the case $k = 0$ holds by definition. Suppose $R_k = \{X : X \leq_T \emptyset^{(k)}\}$. A sequence is uniformly in R_k exactly when its generating function satisfies $g \leq_T \emptyset^{(k)}$. By Lemma 3 with $A = \emptyset^{(k)}$, the limits of such sequences are exactly the sets $X \leq_T \emptyset^{(k+1)}$. Strictness holds because $\emptyset^{(k+1)} \in R_{k+1}$ but, by the relativised halting theorem, $\emptyset^{(k+1)} \notin R_k$. ■

Corollary 2. $R_{k+1} = R_k \cup \partial R_k$. *The boundary of each rung is exactly the new content of the next, and no content of rung $k+1$ arrives from the void of rung k .*

Proposition 1 (Uncertified convergence generates the ladder). *Let $R = \{X : X \leq_T A\}$ for some A . Define the certified closure $\text{cl}_{\text{mod}}(R)$ as the class of limits of sequences uniformly in R that come with a modulus $m \in R$ such that $g(n, s) = X(n)$ for all $s \geq m(n)$. Then $\text{cl}_{\text{mod}}(R) = R$.*

Proof. $X(n) = g(n, m(n))$, which is computable from $g \oplus m \leq_T A$. ■

Remark 2 (A certificate is not the limit). Proposition 1 proves that a generator with a modulus computes the limit. The converse fails, and the failure is the content of the notion: let $g(n, s) = 1$ if machine n halts for the first time at stage s and 0 otherwise. Then g is computable and $\lim_s g(n, s) = 0$ for every n , so the limit is the computable zero set; but a computable modulus m with $g(n, s) = 0$ for $s \geq m(n)$ would decide the halting problem, by running machine n for $m(n)$ steps. Computability of a limit and computability of its certificate are different things, and the ladder is built from the difference.

Topological closure is idempotent, so ordinary spaces carry no ladder. Effective closure is not: $\text{cl}(\text{cl}(R_k)) = R_{k+2} \neq R_{k+1}$. Proposition 1 shows that this failure of idempotence comes entirely from the absence of a certificate. *Rungs are generated by uncertified convergence, and by nothing else in this setting.* An agent that accepts only certified convergence never leaves its rung.

Two natural variants of the mode of approach might be expected to produce intermediate rungs. Neither does.

Proposition 2 (Bounded mind changes generate no intermediate rungs). *For a Turing ideal R let $\text{cl}_1(R)$ be the class of limits of sequences uniformly in R that change their value at each n at most once. Then the Turing ideal generated by $\text{cl}_1(R)$ is $\text{cl}(R)$.*

Proof. For $A \in R$, $A' = \lim_s A'_s$ with A'_s as in the proof of Lemma 3: each value changes at most once, from 0 to 1. So $A' \in \text{cl}_1(R)$ for every $A \in R$, and by Lemma 1 the ideal generated by these jumps is $\text{cl}(R)$. The reverse inclusion $\text{cl}_1(R) \subseteq \text{cl}(R)$ is immediate. ■

The Ershov hierarchy therefore grades *targets* within a boundary, not *spaces*: once spaces are required to be Turing ideals, falsificationist convergence (one mind change) and unbounded trial-and-error reach the same rung. In the terms of Section 10, Popper and Putnam coincide at the level of cognitive spaces and differ only in the fine structure of the boundary.

Proposition 3 (Probabilistic convergence generates no rungs). *Let $R = \{X : X \leq_T A\}$. Say that X is probabilistically approachable from R if the class of $Z \in 2^{\mathbb{N}}$ such that $X \leq_T (A \oplus Z)'$ has positive Lebesgue measure: with positive probability, a random seed lets an R -agent identify X in the limit. Then $X \in \text{cl}(R)$.*

Proof. Almost every Z is Martin-Löf random relative to A' , and for such Z one has $(A \oplus Z)' \equiv_T A' \oplus Z$ [17]. So $X \leq_T A' \oplus Z$ for a positive-measure class of Z , and by the theorem of Sacks and of de Leeuw et al. relativised to A' [30, 9], $X \leq_T A'$. ■

Randomised approximation thus buys no rungs: with probability one a random seed leaves the closure unchanged. (A random seed can, for a measure-zero set of seeds, compute anything at all [18, 13]; see the remarks after Proposition 9.) The ladder is robust under both variations, which is some evidence that the uniform, uncertified definition is the right one and not merely the convenient one.

Theorem 2 (Stall Theorem). *Let $\mathcal{A} = \bigcup_k R_k$ be the arithmetical sets. Then*

$$\text{cl}(\mathcal{A}) = \mathcal{A} \quad \text{and} \quad \emptyset^{(\omega)} \notin \mathcal{A}.$$

Proof. If (X_s) is uniformly in $\mathcal{A}$, its generating function lies in some R_k , so $g \leq_T \emptyset^{(k)}$ and, by Lemma 3, $\lim_s X_s \leq_T \emptyset^{(k+1)}$, which is arithmetical. For the second claim, suppose $\emptyset^{(\omega)} \leq_T \emptyset^{(k)}$. Since $\emptyset^{(k+1)}$ is a column of $\emptyset^{(\omega)}$, we would have $\emptyset^{(k+1)} \leq_T \emptyset^{(\omega)} \leq_T \emptyset^{(k)}$, contradicting the relativised halting theorem. ■

Remark 3 (The climbing sequence). $\emptyset^{(\omega)}$ is nonetheless a limit of arithmetical sets. Let $Y_s = \bigoplus_{k < s} \emptyset^{(k)}$. Each Y_s lies on a finite rung, and $Y_s \rightarrow \emptyset^{(\omega)}$. The sequence is not uniformly in $\mathcal{A}$, however: generating it requires every rung at once. Arithmetic truth is thus approached through steps each of which is reachable, while no reachable agent can generate the approach. It lies in the non-effective closure of $\mathcal{A}$ and in the void of the effective one (Figure 2).

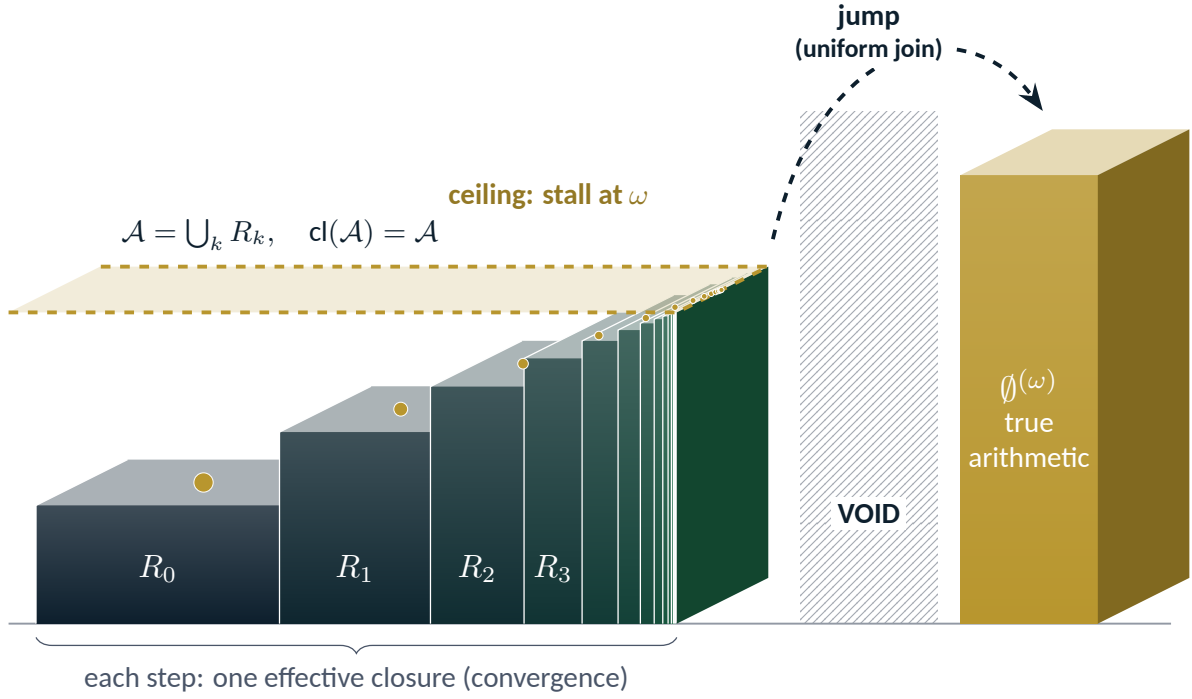


Figure 2: The stall at ω , drawn as a staircase (proportions illustrative). Each step is a rung R_k , and each rise is one application of effective closure. Infinitely many steps fit below the dashed ceiling: the arithmetical sets $\mathcal{A}$ form a fixed point, so no convergent process climbs higher (Theorem 2). Gold points mark the climbing sequence of Remark 3: every point is reachable, but no single rung can generate the whole sequence. True arithmetic, $\emptyset^{(\omega)}$, stands beyond a void and is reached only by a jump, the uniform join of all rungs.

5 JUMPS AND THE UNDECIDABILITY OF JUMPING

Definition 3 (Traversal and jump). A traversal is a sequence $(C_t)_{t \in \mathbb{N}}$ of classes. The step from C_t to C_{t+1} is *continuous* if $C_{t+1} \subseteq \text{cl}(C_t)$ and a *jump* otherwise.

By Theorem 2, any step from $\mathcal{A}$ to a class containing $\emptyset^{(\omega)}$ is a jump. The canonical such step is the *uniform join*, which integrates all finite rungs into a single oracle. A second species is *oracle access*, discussed in Section 10.

Remark 4 (Two words called “jump”). The Turing jump $A \mapsto A'$ is *not* a jump in the sense of the definition above: the step from the ideal below A to the ideal below A' is continuous, since $A' \in \text{cl}(\{X : X \leq_T A\})$ by Lemma 3. The Rung Theorem is precisely the statement that Turing jumps are climbed by convergence. Likewise *axiom adoption*, passing from a theory T to $T + \text{Con}(T)$, leaves the Turing degree unchanged (both theories are computably enumerable) and is therefore continuous here, although it climbs in proof-theoretic strength; the relation between the two axes is taken up in Section 9 and Open issue 5. Where the distinction matters we say “leap” for a discontinuous step and reserve “the jump of A ” for A' .

Proposition 4 (Jumping is undecidable). *There is no algorithm that, given the description of a computable traversal, decides whether it ever jumps.*

Proof. Given a machine M , define the traversal P_M by $C_t = R_0$ if M has not halted within t steps and $C_t = R_2$ otherwise. The map $M \mapsto P_M$ is computable. Since $\text{cl}(R_0) = R_1$ and $R_2 \not\subseteq R_1$ (Theorem 1), P_M contains a jump if and only if M halts. A decision procedure for jumping would therefore decide the halting problem. ■

The proof is elementary, but the consequence is not: a system whose rules are completely known on a lower rung can land outside the closure of its designers’ space, and the designers cannot in general decide in advance whether it will.

5.1 Indistinguishability from below

Proposition 4 concerns a process's rules. We now ask whether a jump can be recognised from a process's behaviour. An *observer* is any function $\Psi : 2^{<\mathbb{N}} \rightarrow \{0, 1\}$, with no computability restriction, which reads finite observations and outputs a verdict. An observer *verifies a class V in the limit* if, for every X , $\Psi(X \upharpoonright s) = 1$ for all sufficiently large s exactly when $X \in V$; it *decides V in the limit* if in addition $\Psi(X \upharpoonright s) = 0$ for all sufficiently large s when $X \notin V$. For a process (X_s) we consider *transparent* processes $X_s = X \upharpoonright s$ (padded with zeros), whose observation up to stage s reveals exactly the prefix $X \upharpoonright s$.

Theorem 3 (Indistinguishability Theorem). *Let C be a countable Turing ideal, for instance any rung R_k or $\mathcal{A}$ itself, and let $V = 2^{\mathbb{N}} \setminus \text{cl}(C)$ be its void.*

1. *Every finite observation is consistent both with a limit in $\text{cl}(C)$ and with a limit in V .*
2. *No observer, of any computational power, verifies V in the limit. In particular none decides $\text{cl}(C)$ in the limit.*
3. *For $C = R_k$, there is an observer computable from $\emptyset^{(k+2)}$ that verifies $\text{cl}(C) = R_{k+1}$ in the limit.*

Proof. (1) $\text{cl}(C)$ is countable, so V is co-countable and dense; $\text{cl}(C)$ contains every finite set and so is dense.

(2) For any Ψ , the set $\{X : \exists s_0 \forall s \geq s_0 \Psi(X \upharpoonright s) = 1\}$ is a countable union of closed sets, since each condition $\Psi(X \upharpoonright s) = 1$ defines a clopen set. If it equalled V , then $\text{cl}(C)$ would be a countable intersection of open sets. A countable dense G_δ subset of Cantor space would be comeagre, but a countable subset of a space without isolated points is meagre, contradicting the Baire category theorem. Deciding $\text{cl}(C)$ in the limit would in particular verify V in the limit.

(3) The observer maintains a current index e , starting at 0. At stage s it uses $\emptyset^{(k+2)}$ to test, for each $n < s$, whether $\Phi_e^{\emptyset^{(k+1)}}(n)$ converges to $X(n)$. If some test fails it moves to $e+1$ and outputs 0; otherwise it outputs 1. If $X \in R_{k+1}$, some correct index exists, every earlier index is eventually refuted, and the output is eventually 1 forever. If $X \notin R_{k+1}$, every index is eventually refuted and the output is 0 infinitely often. ■

The proof of (2) is the topological characterisation of limit verifiability familiar from formal learning theory [19]; the content lies in the statement. Note that (2) uses only that $\text{cl}(C)$ is countable and dense: it holds for any countable C containing the computable sets, whether or not C is an ideal, and for observers relativised to any oracle whatever. Three consequences are worth recording.

- *Crossing the stall is invisible.* Taking $C = \mathcal{A}$: no observer can recognise, even in the limit, that an observed process has passed the ω -stall.
- *Rules and behaviour.* With Proposition 4, a jump can be recognised neither from a process's rules nor from its behaviour.
- *What a void is.* Content on the next rung is recognisable in the limit, although only from above (part 3). A void is not recognisable as a void even in the limit, by any observer (part 2). Seen from below, the two are indistinguishable at every finite stage, and only one of them ever resolves.

Two limitations should be stated. The theorem concerns observers without access to the generating rules: if an observer knows that a process is generated at its own rung, the limit lies in the next rung automatically. And we have not determined whether an R_k -agent can itself verify membership of R_{k+1} in the limit; part (3) uses an oracle two rungs above it.

6 AGENTS: PROMOTION, TRANSMISSION AND CERTIFICATION

The Rung Theorem describes classes, not agents. An agent at R_k can generate approximations converging to any set in R_{k+1} , but it cannot tell when they have stabilised (Proposition 1). It is therefore not obvious who, if anyone, turns a limit-computable approximation into content that can be used as an oracle for the next rung. This section answers that question, and in doing so formalises the agent's capacities to receive, reproduce, teach and certify.

6.1 Three relations and promotion

Definition 4 (Agents; settling, identifying, holding). An *agent* is a pair (C, P) of Turing ideals with $C \subseteq P$. C is its *certified space* and P its *committed space*: P is the ideal generated by C together with the sets the agent has promoted (below). Let $X \subseteq \mathbb{N}$.

1. The agent *settles* X if $X \in C$.
2. The agent *identifies* X in the limit if $X \in \text{cl}(P)$, that is, if it can generate approximations converging to X , using its commitments as oracle answers.
3. The agent *holds* X as an oracle if $X \in P$.

An agent with $C = P$ is *uncommitted*.

Remark 5 (Agents are individuated by their spaces). An agent is the pair (C, P) and nothing else. Two agents with the same pair are, in this model, the same agent: there is no further attribute to distinguish them. Consequently a *population* of agents is a set of pairs, not a multiset, and “how many agents occupy a space” is not a question the model can pose. Section 8 builds on this.

For an uncommitted agent, holding is settling, and for $X \in \partial C$ the agent identifies X in the limit but neither settles nor holds it. The gap between the second and third relations is exactly the missing operation.

Definition 5 (Promotion). A *promotion* replaces P by the ideal generated by $P \cup \{X\}$ for some $X \in \partial P$, leaving C unchanged. The agent commits to its current approximations to X as though they were final, and uses them as oracle answers in all further computation, revising dependent conclusions whenever an approximation changes. The *depth* of an agent is the least number of promotions that generate P from C .

A promotion is a continuous step in the sense of Section 5, since the new content lies in $\text{cl}(P)$. It is also uncertified: at no finite stage can the agent verify that its current approximation to X is correct.

Remark 6 (What promotion changes). A promoting agent remains, as a process, exactly what it was: its state at stage s is computed by its rules from s and its certified oracle. Promotion changes nothing about what the agent computes and everything about which sets its beliefs converge to. “Holding X as an oracle” therefore means *believing X , correctly in the limit*, and the committed space P is the space of the agent’s limiting beliefs. The next proposition makes this exact: an agent of depth k is a computable process whose beliefs are k -fold limits.

Proposition 5 (Climbing is nested promotion). For every k and every X : $X \in R_k$ if and only if there is a computable g with

$$X(n) = \lim_{s_1 \rightarrow \infty} \lim_{s_2 \rightarrow \infty} \cdots \lim_{s_k \rightarrow \infty} g(n, s_1, \dots, s_k) \quad \text{for all } n.$$

Consequently, an agent with certified space R_0 and depth k holds at most R_k and identifies at most R_{k+1} in the limit, with equality when it promotes the canonical sets $\emptyset', \dots, \emptyset^{(k)}$; every set it relies upon at depth k rests on k nested uncertified commitments.

Proof. Induction on k using Lemma 3 relativised to $\emptyset^{(k-1)}$, which converts one limit into one jump. ■

This resolves the question of who performs the conversion. No agent converts a limit into certified content. The ladder of Theorem 1 is climbed by promotion, a commitment that is correct in the limit and never verified along the way. Certified content can only arrive from outside, by one of the routes below. Every canonical climb therefore contains a small leap of commitment, in keeping with the general principle that new rungs are reached only uncertified.

6.2 Transmission

Proposition 6 (What can be transmitted). Let a recipient have cognitive space C .

1. Content $X \in C$ can be transmitted as an algorithm, which the recipient can run and check.

2. Content $X \in \partial C$ can be transmitted as a method: a generator $g \in C$ with $\lim_s g(\cdot, s) = X$. The recipient then identifies X in the limit, and holds it only after its own promotion.
3. Content $X \notin \text{cl}(C)$ admits no method at the recipient's rung. It can be transmitted only as finite fragments (every finite initial segment is computable), as testimony (answers supplied from outside), or as an invitation to adopt it by a leap.

Proof. Items (1) and (2) are the definitions of C and $\text{cl}(C)$. For (3), a method would be a generator in C converging to X , placing X in $\text{cl}(C)$. ■

Transmission is therefore asymmetric. Content passes upward freely, since a higher ideal contains a lower one. Content acquired by a jump cannot pass back down through the threshold of the rung as method, only as fragments, testimony or invitation.

6.3 Teaching, mastery, certification and examination

Fix a framework F : a theory or a class of valid outputs, computable from a set f . We define roles by what an agent can do, not by what it knows.

Definition 6 (Roles). 1. An agent is *competent* in F if it settles f : it can generate and check instances of F .

2. An agent is a *master* of F if it can make other agents competent in F , which in this model means it can mark arbitrary attempts reliably.
3. An agent is a *certifier* for F if it settles the consistency of every f -computable theory.
4. An agent is an *examiner* for F if it settles whether an arbitrary student's generator always outputs valid members of F .

Proposition 7 (The cost of each role). *Let the framework be computable from f .*

1. Competence requires f . Mastery of a decidable framework also requires only f . Mastery of a generative framework (a c.e. theory T) marked on bare claims requires the degree of T , which is Σ_1^0 relative to f and so at most f' , with f' attained when T is creative relative to f ; if students must show their working, marking is decidable and f suffices again.
2. Certification requires f' and nothing less: the set CONS_f of indices of consistent f -computable theories is $\Pi_1^0(f)$ -complete, hence of degree f' .
3. Examination requires f'' , since totality with validity is Π_2^0 relative to f . It is identifiable in the limit from f' with unbounded mind changes, and not identifiable in the limit from f at all.
4. Each assurance is limit-identifiable one rung below the rung at which it is certain. Certification is identifiable in the limit from f with at most one mind change: believe consistency until a contradiction appears.

Proof. The quantifier counts follow from the definitions. For (2): inconsistency of an f -c.e. theory is witnessed by a finite derivation, so CONS_f is $\Pi_1^0(f)$ and decidable in f' . For hardness, given an f -oracle machine e , uniformly build the theory T_e^f that is a fixed consistent base theory unless $\Phi_e^f(e)$ halts, at which stage it enumerates a contradiction; then $e \notin K^f$ iff T_e^f is consistent, so K^f many-one reduces to the complement of CONS_f . No appeal to relativised incompleteness is needed: the statement is an index-set completeness theorem. The limit statements are Lemma 3 applied to Σ_1^0 and Π_2^0 questions, together with the fact that a Π_2^0 -complete question relative to f is not Δ_2^0 relative to f . ■

Item (4) has a familiar reading: the one-mind-change shadow of certification is falsificationism. The role ladder and the Ershov grading of mind changes (none, one, unbounded) are the same structure seen from two sides; by Proposition 2 the grading distinguishes targets within a boundary, never rungs.

Proposition 8 (Marking is the channel of transmission). *Suppose a student learns F in the limit from a teacher, choosing among a uniformly decidable family of candidate frameworks. If the student sees only valid examples, no family containing all finite frameworks and at least one infinite framework is learnable in this way. If the teacher marks the student's attempts, every uniformly decidable family is learnable in the limit [14].*

Teaching therefore transmits competence as method, through correction rather than example, and the competence it produces is itself reached in the limit: the student never knows the moment it has learned.

Proposition 9 (Practice and noise). **1.** *If an agent is competent in F , the canonical promotion (of its own jump) gives it the resources of a certifier: $a \geq_T f$ implies $a' \geq_T f'$. Promotion of an arbitrary boundary set need not: promoting a low set adds nothing at the next rung.*

2. *Randomness alone confers no new computational power: a set computable from a positive-measure class of oracles is computable [30, 9].*

Proof. (1) is monotonicity of the jump. (2) is the cited theorem. ■

Ascent therefore comes from promotion, which in an evolutionary setting is variation filtered by selection over time, and not from noise. Noise diversifies agents; selection lets them climb. The precise form of this slogan is the pair of Proposition 3 and the Kučera–Gács theorem [18, 13]: every set, $\emptyset^{(\omega)}$ included, is computable from some Martin-Löf random seed, but the seeds that help form a null class, and by Theorem 3(2) no selection process can recognise, even in the limit, that it has drawn one.

Improvement. Two kinds of change follow. *Certified evolution* modifies a framework within the rung below the agent: a certifier at f' can build and certify arbitrary variants of f -computable frameworks, and the results are teachable to competent agents at f . *Uncertified ascent* is promotion or a leap, and alone reaches a new rung. So every agent is a master of the rung below and a student of its own, and no agent reaches a higher rung with a certificate in hand.

7 THE GEOMETRY OF POSITIONS: A SUMMARY

The agent model of Section 6 describes what a single agent can do. To compare two agents we need a distance, and the companion paper [4] studies one. For sets x, y let $\delta(x, y)$ be the least k with $y \leq_T x^{(k)}$, the number of closure steps (Theorem 1) an agent at x needs before y lies in its space, and let $d(x, y) = \max(\delta(x, y), \delta(y, x))$, the lag before the two are mutually intelligible. Write $h(x) = d(x, \emptyset)$. We record the results we use, with proofs in [4].

Theorem 4 (Geometry of the rung distance; summary of [4]). **1.** *d is an extended metric on Turing degrees, finite exactly between mutually arithmetical sets. Its components are the classes of mutually arithmetical sets; the component of $\emptyset$ is $\mathcal{A}$.*

- 2.** *Every canonical ascent $x, x', x'', \dots$ is a geodesic ray. All such rays are asymptotic to the trunk $\emptyset, \emptyset', \dots$ and determine a single ideal point ξ , which we identify with true arithmetic; its Busemann function records the high side of the high/low hierarchy. A canonical ascent can fellow-travel the trunk at distance exactly one for ever without merging into it (intermediate c.e. degrees).*
- 3.** *Every geodesic ray r has $h(r(i)) = i + O(1)$: no ray outruns the trunk in height. The boundary at infinity is a single point if and only if every ray recovers the canonical content of its own level with bounded lag; this is open.*
- 4.** *There is an ultrametric $D(x, y) = \min\{k : x \oplus \emptyset^{(k)} \equiv_T y \oplus \emptyset^{(k)}\}$ whose nested balls form a rooted tree (Rook's Tree, Figure 3), and an excess $\varepsilon(x)$, zero exactly on the generalised-low agents, with $d \leq D \leq d + \max(\varepsilon(x), \varepsilon(y))$. Agents of bounded excess form a Gromov-hyperbolic family.*
- 5.** *On two mutually random generalised-low agents a, b of height H and their educated versions $a \oplus \emptyset^{(s)}, b \oplus \emptyset^{(t)}$, the distance is $\max(s, t, H) - \min(s, t)$: the longer leg of the route through the tree, not its*

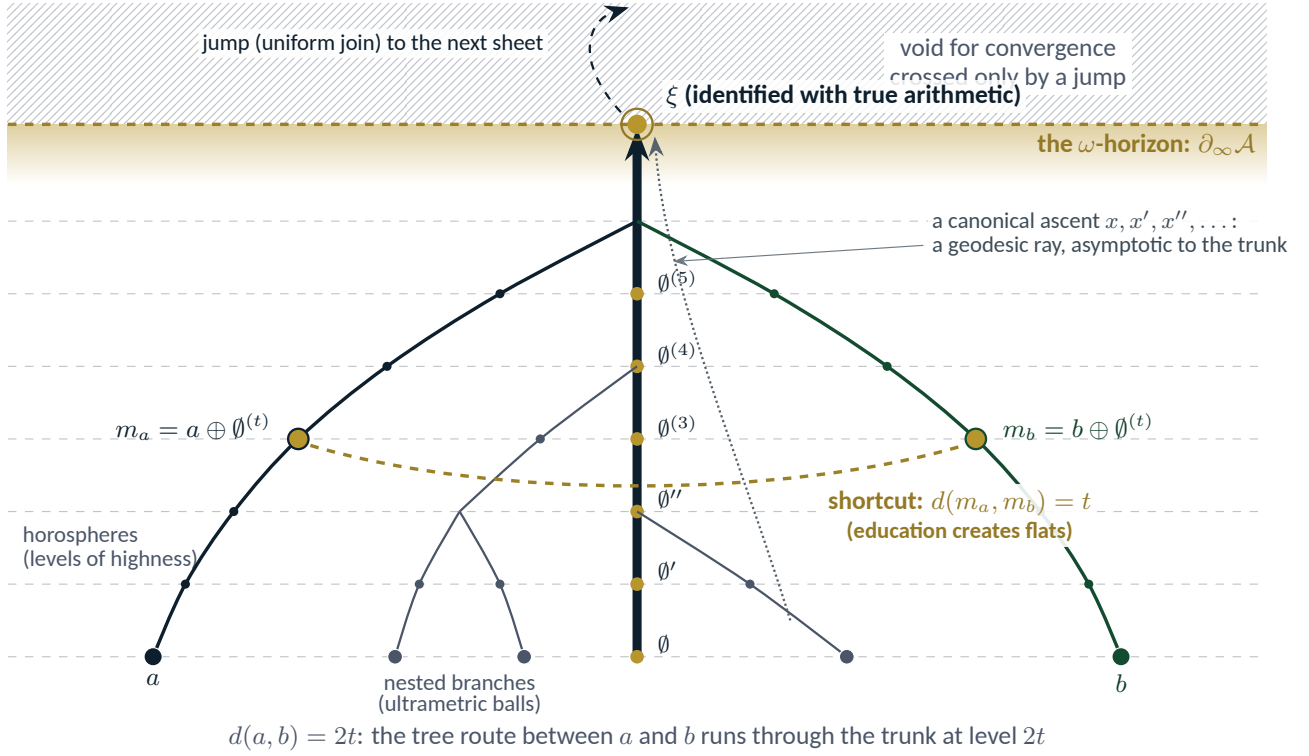


Figure 3: Rook's Tree (schematic; horizontal positions are illustrative). The trunk $\emptyset, \emptyset', \emptyset'', \dots$ rises towards the ideal point ξ , identified with true arithmetic, on the ω -horizon $\partial_\infty \mathcal{A}$. Dashed horizontal lines are horospheres centred at ξ , measuring highness. Branches are chains of D -balls (Theorem 4(4)): the vertex on the branch of x at level k is the ball $[x]_k$, labelled by $x \oplus \emptyset^{(k)}$, and each branch joins the trunk at height $h(x)$; generalised-low agents are leaves on which d agrees with the tree. Branches that merge below the trunk form nested ultrametric balls. The gold points m_a, m_b are two random agents after absorbing the canonical content $\emptyset^{(t)}$: they sit at distance t , although the route through the tree would place them $2t$ apart, which is the flat of Theorem 4(5). Above the horizon lies the void for convergence, crossed only by a jump onto the next sheet.

length. Consequently $(\mathcal{A}, d)$ is 2-bi-Lipschitz to a tree on this family yet fails the four-point condition with unbounded defect, so it is not Gromov-hyperbolic in the four-point sense. It is not a graph metric: for $H \geq 4$, no neighbour of a is closer to b than a is (a dead end). Whether it is quasi-isometric to a tree is open; [4] reduces the question to Manning's bottleneck property for the graph of unit steps and refutes tree-likeness conditionally, under an embedding hypothesis for countable jump partial orders with least element.

Three readings of the figure matter for this paper (an image of the tree, not an argument, is given as a postscript). First, *within a component, separation is a lag, not a void*: any two arithmetical agents become mutually intelligible after finitely many closure steps. Permanent voids exist only between components, that is, between agents that are not arithmetical in one another. Second, *education creates flats*: two random agents at distance $2t$ that each absorb the canonical content $\emptyset^{(t)}$ end up at distance t , a shortcut no tree admits, and this is the whole obstruction to hyperbolicity. Third, the boundary question (item 3) is a question about lag: a mind that keeps pace with the canonical ladder in height but is slow to recover its content is the only kind of mind that could face a second horizon.

8 ENCOUNTERS: ONE APPLICATION OF THE FRAMEWORK

The agent model describes one agent; the rung distance compares two. This section asks what happens when agents meet. It is included as a worked example of what the framework generates; the population dynamics, the cosmological reading and the strategic analysis that follow from it are developed in the sequel [2] and are attached at the branching points listed in Section 12. Throughout, agents are principal: an agent is identified

with a degree x , its certified space is the ideal below x , and by Remark 5 a population is a finite set $\mathcal{P}$ of degrees.

8.1 Three outcomes

Definition 7 (Encounter). An encounter between agents $x, y \in \mathcal{P}$ has one of three outcomes.

1. *Merger*: the two pool as one machine with both oracles. $\mathcal{P} \mapsto (\mathcal{P} \setminus \{x, y\}) \cup \{x \oplus y\}$, and $x \oplus y$ is settled by the new agent.
2. *Elimination*: $\mathcal{P} \mapsto \mathcal{P} \setminus \{y\}$. If the survivor absorbs the eliminated agent's content the outcome is a merger.
3. *Trade*: each agent promotes some of the other's content, received as testimony (Proposition 6). x 's committed space becomes the ideal below $x \oplus z$ for some $z \leq_T y$, and symmetrically. The population keeps two members.

Merger and elimination-with-absorption are the same operation on $\mathcal{P}$; cooperation and conquest are not distinguished by the model. Because the model records no history, the new member $x \oplus y$ is not marked as the descendant of x and y : merger is the elimination of both parties and the creation of one. What the model does retain, for as long as two agents remain distinct, is the lag $d(x, y)$ between them, which is an invariant of the pair and not a memory.

Proposition 10 (Mergers are lateral). If $x, y \leq_T \emptyset^{(k)}$ then $x \oplus y \leq_T \emptyset^{(k)}$. No sequence of mergers leaves a rung; only promotion climbs.

Proposition 11 (Merger forgets). The join does not determine its parts: for every z that is a join of two incomparable degrees, and every such decomposition $z = x \oplus y$, there are $y' \not\equiv_T y$ with $z \equiv_T x \oplus y'$. Given a merged agent and one of its parts, the other is not recoverable.

Proof. Take $y' = y \oplus w$ for any $w \leq_T x$ with $w \not\leq_T y$; such w exists since $x \not\leq_T y$. ■

Theorem 5 (Merger creates certified content unavailable to either constituent). There are agents $x, y <_T \emptyset'$, neither of which settles $\emptyset'$ and each of which identifies $\emptyset'$ only in the limit, whose merger settles $\emptyset'$. More generally, every noncomputable c.e. space is the merger of two strictly smaller c.e. spaces, neither of which subsumes the other.

Proof. Sacks's splitting theorem [26]: every noncomputable c.e. degree is the join of two incomparable c.e. degrees below it. Apply it to $\emptyset'$. Each part x satisfies $\emptyset' \leq_T x'$ (any c.e. set is Δ_2^0 relative to anything), so each identifies $\emptyset'$ in the limit; neither settles it. ■

The theorem gives merger a status no other operation has: it converts boundary content into interior content with a certificate. Promotion reaches the same content only as belief (Remark 6), and merger is not promotion in the sense of Definition 5: the constituents disappear and a new agent exists. Merger is the one certified route to new content inside a rung.

Proposition 12 (Irreducible agents exist). There are agents below $\emptyset'$ whose space is not the merger of any two strictly smaller noncomputable spaces.

Proof. A minimal degree is not the join of two lesser noncomputable degrees, and minimal degrees exist below $\emptyset'$ [27]. ■

Proposition 13 (Shared ground need not be a space). The overlap $C_x \cap C_y$ of two agents' spaces is a Turing ideal but need not be principal: there are pairs x, y such that no single agent's space is exactly their common ground.

Proof. The Turing degrees are not a lattice: there are pairs with no greatest lower bound [28, 33]. For such a pair the intersection ideal has no greatest element. ■

Proposition 14 (Trade costs lag; merger is free). *Agent x can hold y as a committed oracle after k promotions if and only if $\delta(x, y) \leq k$. Mutual absorption by trade therefore costs $d(x, y)$ rounds of uncertified promotion, and even then the two certified spaces remain distinct. Merger costs no promotion and certifies $x \oplus y$.*

Proof. After k promotions the committed space is at most the ideal below $x^{(k)}$ (Proposition 5), and y lies in it iff $y \leq_T x^{(k)}$; the canonical promotions attain the bound. ■

This gives the rung distance an operational meaning it did not have in the companion paper: $d(x, y)$ is the cost of union by trade, in uncertified commitments, against a merger cost of zero. That merger costs nothing is a convention of the model, not a consequence of the degrees: pooling two agents into one machine may in practice be far more expensive than teaching, and the strategic readings in [2] inherit the convention.

8.2 Individuality is unverifiable

Can an agent tell whether a peer is a copy of itself? A peer's content arrives as testimony, prefix by prefix, so the question is one of limit verification in the sense of Section 5. Let $I_x = \{y : y \leq_T x\}$ be the set of peers within x 's space.

Theorem 6 (Subsumption is verifiable from above; non-subsumption by no one). **1.** *There is an observer computable from x' that verifies I_x in the limit: an agent that has promoted its own jump can come to believe, correctly in the limit, that a peer lies within its space.*

2. *No observer, of any computational power, verifies $2^{\mathbb{N}} \setminus I_x$ in the limit: no agent can ever come to believe, by any procedure, that a peer exceeds its space.*

Proof. (1) The observer holds a current index e and at stage s uses x' to check, for each $n < s$, that $\Phi_e^x(n)$ converges and equals $y(n)$; on failure it moves to $e+1$ and outputs 0, otherwise it outputs 1. If $y \in I_x$ a correct index exists, every earlier index is refuted at a finite stage, and the output is eventually 1 for ever; if $y \notin I_x$ every index is refuted and the output is 0 infinitely often. (2) I_x is countable and dense. If its complement were verifiable in the limit it would be a countable union of closed sets, making I_x a dense G_δ , which is impossible for a countable set (as in the proof of Theorem 3(2)). ■

The asymmetry is the one the dark-forest argument needs and never states. From passive testimony an agent's belief that a peer lies within its space converges correctly (after one promotion); no belief that a peer has content outside its space can converge correctly, by any procedure. In the terms used throughout the series: *subsumption* of a peer is limit-verifiable, its *strict superiority* is not. (The sequel [2] adds an interactive protocol by which the subsumed party can also come to believe it is subsumed; even then it cannot tell equivalence from strict superiority.) Combined with Proposition 11, a merged agent cannot tell whether it is the product of a merger, of a conquest, or of neither, and cannot tell whether the peers it meets are fragments of what it once was. Individuality is unrecoverable in both directions.

9 TRANSFINITE EXTENSION AND THE EFFECTIVE CEILING

The results of this section are classical and are cited rather than reproved [20, 34, 29]. Allowing uniform joins at limit stages continues the ladder beyond ω . Limits must be indexed by ordinals the agent can name, so the ladder is indexed by Kleene's system of notations $\mathcal{O}$. For each notation $a \in \mathcal{O}$ the set H_a is defined by the jump at successors and the uniform join along fundamental sequences at limits. By Spector's theorem, the degree of H_a depends only on the ordinal denoted.

The sets computable from some H_a are the hyperarithmetical sets HYP, which coincide with the Δ_1^1 sets. HYP is closed under the jump and hence under effective closure: if $g \leq_T H_a$ then $\lim g \leq_T H'_a$. The ladder therefore runs through every ordinal below the Church-Kleene ordinal ω_1^{CK} and no further by effective means. Kleene's $\mathcal{O}$ is Π_1^1 -complete and not hyperarithmetical: climbing past ω_1^{CK} would require certifying that a proposed notation is well-founded, and that judgement is not effective. This is the obstruction Turing met in his ordinal logics [36]. The proof-theoretic axis of Remark 4 runs into the same wall. Feferman's transfinite recursive progressions iterate reflection principles along notations in $\mathcal{O}$; a progression based on uniform

reflection proves every true arithmetic sentence along *some* path through $\mathcal{O}$, but the path is not effectively recognisable [12]. Axiom adoption thus has the same ceiling as convergence (true arithmetic, reachable only along a notation the agent cannot certify) and the same obstruction (the Π_1^1 -completeness of $\mathcal{O}$). The two axes are different orderings with the same horizon.

Stage	Class reached	Operation required	Status for an effective agent
Rung $k < \omega$	Δ_{k+1}^0	k closures	Reachable by convergence
Stall at ω	Arithmetical sets	Closure now idempotent	Ceiling for convergence (in this model)
Past ω	$\emptyset^{(\omega)}$, true arithmetic	Uniform join (a jump)	Reachable only by jump
Below ω_1^{CK}	$HYP = \Delta_1^1$	Closures and joins along notations	Reachable by effective traversal
From ω_1^{CK}	$\mathcal{O}$ and beyond	Non-effective well-foundedness	Void for every effective agent

10 INTERPRETATION

If the postulates hold, the results translate into claims about minds, horizons and traversal. We state these as interpretations, not theorems.

Four regions. Relative to an agent, a target is feasible or costly if it lies in the interior, limit-approachable if it lies on the boundary, and void if it lies outside the effective closure. The last distinction is membership in $cl(C)$, not a cost threshold, and within the boundary the Ershov hierarchy grades difficulty by mind changes, without ever producing a rung of its own (Proposition 2). By Corollary 1 the void is the generic case.

Rungs and epistemic tolerance. Proposition 1 orders rungs by what an agent accepts in place of proof. An agent that accepts only proofs stays on its rung. An agent that accepts stabilising conjecture, the method of empirical science on Putnam’s and Gold’s analysis [25, 15], climbs, and by Proposition 5 it does so by promotion, never by certification.

Convergent theories. Remark 3 gives a precise form to the question of whether a sequence of improving physical theories converges on the truth. A sequence of reachable theories can converge to a limit that no single rung can generate. From inside the sequence, and by Theorem 3(2) from any vantage whatever, a limit on the next rung (a limit-approachable frontier) and a limit requiring a jump (a void) are indistinguishable at every finite stage.

Undetectability from below. Theorem 3 replaces an argument, made in an earlier version of this work, by analogy with Chaitin’s incompleteness theorem [7]. A cognitive void is exactly what cannot be recognised as a void, by any observer and even in the limit, while the next rung can be recognised from above. A squirrel’s inability to perceive algebra as missing is, on this reading, not a failure of effort but a theorem about observation.

Physical limits separate existence, approach and certification as effective limits do. Two results from the physics of lattice Hamiltonians illustrate Propositions 1 and 2 rather than extend them. A spectral gap $\Delta > 0$ supplies a certificate of convergence when the approximation dynamics admits an error bound in terms of it (imaginary-time projection converges to the ground state with an explicit rate set by Δ and the initial overlap); a gapless system converges, if at all, without such a bound. Under that conditional reading, the theorem of Cubitt, Pérez-García and Wolf, that whether a translationally invariant Hamiltonian is gapped is undecidable even though each instance is definitely one or the other [8], says that membership of the certified regime can itself be undecidable from the rules; this is the physical counterpart of Remark 2, that a certificate carries information the limit does not. The size-driven transitions of Bausch et al. [5], in which the gap stays constant while the phase changes once, at a lattice size that grows faster than every computable function of the local

dimension, are a one-mind-change process whose change point has no computable bound: finite-size evidence for the phase is a promotion in the sense of Definition 5, and Proposition 2 says that a single mind change buys no certificate. Neither result is used anywhere in this paper; the sequel [2] develops the correspondence.

Universal agents and what separates them. Postulate 1 has a consequence that should be stated plainly. Under unbounded time with a fixed architecture, every Turing-universal agent without an oracle has cognitive space R_0 : humans, machines and any universal animal share a single space, and the ladder separates agents only by oracle or, through Section 6, by commitment depth. The motivating gaps of the introduction are therefore either sub-universal (Open issue 1) or super-Turing (Open issue 4); what the model says about universal agents is the promotion story and the geometry of Section 7. Within $\mathcal{A}$ every distance is finite (Theorem 4(1)), so any two arithmetical agents are mutually intelligible after finitely many rungs: within a component, separation is a lag, not a void. Permanent voids exist only between *components* of d , that is, between agents that are not arithmetical in one another.

The dark forest. Liu's game theory requires players who share a rung, so that each can model the other's options. Section 8 makes the geography exact. Across components there is no common game: one side is arithmetical in the other or neither is, and both relations are definable on degrees. Inside a component every separation is a finite lag, the game exists, and the theory says precisely what the players cannot know before choosing: by Theorem 6, each can come to believe, correctly in the limit, that the other lies within its space, but can never come to believe that the other has content outside it. Liu's chain of suspicion is this asymmetry. Cooperation is then a decision to promote the other's testimony (Proposition 14), or to pool (Theorem 5); conflict is a refusal to; and at the level of spaces conquest-with-absorption and cooperation are the same operation (Definition 7). On this reading the dark forest describes the intra-component case, played on promoted beliefs about the opponent's space, and relations across components are closer to ecology than to strategy. The sequel [2] takes this further: which superiorities can be demonstrated, and what a single non-rational actor does to a population of rational ones.

11 RELATED WORK

The results of Section 4 and of the transfinite section are classical: the limit lemma [32], the arithmetical hierarchy [24], the undefinability of truth [35], the hyperarithmetical hierarchy [20, 34] and ordinal logics [36]. Limit computation as a model of inquiry originates with Putnam and Gold [25, 15] and is developed systematically in Kelly's formal learning theory [19], which is the closest existing framework and the most likely source of overlap. Counting limits is the organising idea of Weihrauch reducibility, which classifies problems by how many applications of the limit operator they require [6]; that programme is the closest mathematical precedent for our closure ladder and should be compared in detail. Grading by mind changes is the Ershov hierarchy [11]; the closure ladder is also effective Baire class in the sense of effective descriptive set theory [23]; and Turing ideals are standard in reverse mathematics and degree theory. Philosophical antecedents include McGinn's cognitive closure [22] and, in opposition, Deutsch's account of universal explainers [10].

The remaining results are, to the author's knowledge, new, although their proofs rest on standard tools: Proposition 4; the Indistinguishability Theorem, whose proof is the topological characterisation of limit verifiability [19]; the agent model; and, in the companion paper [4], the rung distance with its point at infinity, horofunction, Shadow Tree Theorem, non-hyperbolicity theorem and hidden-information theorem. More broadly, and subject to the fuller comparison called for in Open issue 14, particularly with the Weihrauch and learning-theoretic literature: the reading of effective closure as convergent inquiry, with its failure of idempotence as the origin of cognitive rungs; the identification of the ω -stall as the first void for convergence-based cognition in this model, together with the climbing sequence; the agent model of Section 6, with promotion as the mechanism of ascent and the role ladder of competence, mastery, certification and examination; the rung distance and its geometry, summarised in Section 7; and, as interpretation only, the connections to horizons and to cross-rung game theory.

12 BRANCHING POINTS

The core of this paper is Sections 2–6: the postulates, effective closure, the rung ladder, the stall, the Indistinguishability Theorem and the agent model. Everything else attaches to that core at a small number of identifiable points, and can be detached without loss to it. We list them so that a reader can see where the framework ends and its applications begin.

1. **Geometry** [4] attaches at Definition 4 and Theorem 1: the rung distance d counts closure steps between principal agents. Nothing in the core depends on it; Section 7 summarises it because Section 8 uses d as a cost.
2. **Encounters** (Section 8) attach at Remark 5: once agents are individuated by their spaces, a population is a set of degrees and merger, elimination and trade are operations on it. Theorems 5 and 6 use only the core and Sacks splitting.
3. **Population dynamics, cosmic time and finite computation** [2] attach at Proposition 5 (promotion as the only upward move) and at the encounter operations; the resource-bounded attaches at Remark 6, since the committed space at a finite stage is the object a finite budget realises.
4. **Strategy** [2] attaches at Theorem 6 and Proposition 6: what can be demonstrated between agents is fixed by what testimony can carry and what limit verification can decide. The non-rational actor of [2] extends the ontology of Definition 4 to a strategic agent (C, P, r) with a policy source r outside P : a policy that has taken an oracle leap (Section 5) which the agent's cognition has not.
5. **Physics** attaches only at the level of interpretation, as illustration (Section 10); no result depends on it. The physical readings that make claims of their own (horizons as certificates, determinism, the two moduli of spectral convergence) are in [2].

13 PROGRAMMATIC SYNTHESIS: REACHABILITY, GEOMETRY AND CERTIFICATION

The papers of this programme study one object at three levels of description, and a fourth paper [3] applies the third to Fermi's question; it is worth saying once what holds them together, beyond the cross-references. The common question is not what an effective agent can compute, but how effective access changes when four things are kept apart: convergence, certification, distance and interaction.

Three orderings. This paper supplies the order of *reachability*: a cognitive space is a Turing ideal, effective closure adds the limits of approximations generated within it, and iterated closure produces the arithmetical ladder (Theorem 1). The ladder exists because convergence is uncertified: an approximation that comes with a modulus at the same rung has a limit already computable there, and certified closure is idempotent (Proposition 1). A boundary thus separates what can be approached from what can be settled; a void separates what can be approached from what cannot be approached at all. Reachability says that content at x is available at y when $x \leq_T y$, and the jump is its canonical upward motion; it does not say how far apart two positions are, how costly reconciliation is, or how information can stay concealed inside a degree. Paper A [4] introduces the second ordering, which is *geometric*: $\delta(x, y) = \min\{k : y \leq_T x^{(k)}\}$ counts the closure steps before y is reachable from x , and $d = \max(\delta, \delta)$ is the delay before two positions are mutually accessible. Canonical ascent is geodesic, height is distance from the computable base, and the mutually arithmetical degrees form the finite-distance components. Two structures are superposed on this geometry: the ultrametric D , whose nested balls form a shadow tree recording when canonical content suffices to make two degrees equivalent, and the rung metric d itself, which can take shortcuts across that tree, since shared information shortens mutual distance without reproducing the tree route, and whose dead ends show that a distance need not be realisable by unit moves. The geometry is therefore neither hierarchical nor flat: a vertical direction supplied by the jump, a tree-like nesting supplied by equivalence after canonical joins, and lateral compression supplied by shared information. Whether the whole space is coarsely a tree is open, and both natural maps into the shadow tree fail [4, Section 7.3].

The third ordering is implicit here and explicit only across the programme: the ordering of *certification*. A value and the information about when that value has stabilised are different resources. A computable ap-

proximation may converge to a computable limit with no computable modulus; a modulus turns asymptotic convergence into a finite certificate. The informational content of a process is therefore not exhausted by its limit: there is also information in when the limit becomes safe to use. Paper A's hidden-information theorem sharpens this to a statement about degrees. What a degree conceals from the canonical ladder may be not a bit but the delay at which the bit becomes recoverable; generic coding makes those delays escape every rung of domination while the limiting values remain available one jump up, whereas random content, being dominated one jump up, can conceal nothing beyond two rungs. The three orderings do not collapse into one another. Turing reducibility orders content; the rung metric orders the closure steps needed to reconcile content; moduli order the effective availability of certificates. Two agents may hold the same limiting content and differ in whether either holds a usable stabilisation bound; two may be one rung apart while the content hidden in that step needs many canonical rungs to audit; and a process may converge correctly while no agent at its own level can know that it has. The distinction between settling, identifying in the limit and holding a promoted commitment (Definition 4) is the epistemic form of this separation.

Time and strategy. Paper C [2] adds time. Once agents promote at different rates, exchange testimony and meet, the static geometry becomes a dynamics: canonical lineages climb the vertical direction, rate differences produce one-sided lag, and passage into a mutually non-arithmetical component requires an oracle leap rather than further promotion (Section 5). At finite physical time the infinite classification collapses, since every realised history is finite, and Paper C separates the realised trace from the counterfactual question of which of its entries would stay settled were computation continued without bound. That is the finite-time form of the modulus distinction: what has happened, what will converge, and what can be certified to have converged are three objects. Strategy adds one more axis without replacing the others. A strategic agent (C, P, r) may condition its actions on a source r outside its committed space; cognitive subsumption then no longer implies strategic predictability, and prediction, demonstration and audit become three questions: whether a peer can anticipate the agent's next action, whether superiority can be shown through an observable stream, and whether the source can later be reconstructed from the trace. The random/generic contrast connects this layer back to the geometry. Randomness makes superiority demonstrable, through a positive effectively closed class that no inferior agent can enter, yet it is recoverable in hindsight from its effects, with a delay the observer can bound, because random content is dominated one jump up; genericity supplies the freedom to hide information in delays that no finite rung bounds, and demonstrates nothing. Randomness and genericity are thus not merely two sources of incomputability but opposite sides of one distinction, between information that is statistically irreducible yet eventually certifiable and information whose revelation schedule itself resists certification. The physical readings of Paper C sit at this level too: a spectral gap matters not because an energy gap is a jump but because, under suitable dynamics, it yields a computable rate of approach and so a certificate; the undecidability of the gap is then undecidability of the certificate regime, and size-driven transitions separate a dynamical modulus from a size modulus.

One invariant, five horizons. Seen this way the papers are ordered by scale. This one asks what is reachable and certifiable for one agent; Paper A, what geometry the closure steps induce when used as distance; Paper C, what that geometry becomes when agents move through it in time, interact, exhaust finite resources and act on information outside their cognition. The fourth paper [3] is where that dynamics lands: it turns it into numbers for the Fermi problem, separating five filters between a physical population and an observed one (existence, persistence, causal reach, channel access, cognitive access), composing them into an observer-dependent accessible intensity whose exponential is the probability of silence, and finding preliminary numerical evidence for an accessibility percolation governed by the persistence of information channels. The invariant throughout is effective access: to a value, to another agent, to a stabilisation time, or to the source of an action; and the recurring obstruction is not difficulty in any single sense but the absence of a channel: no reduction, no finite sequence of closure steps, no modulus, no reliable testimony, no trace that carries the missing source. The programme's horizons are correspondingly of five different kinds and should not be identified: the ω -stall of closure is a horizon of reachability; the ideal boundary of the rung geometry is a horizon of asymptotic position; the certification horizon separates convergence from effectively bounded convergence; finite cosmological computation makes a horizon of realised history; and the Gambler makes an observer-relative horizon of pre-

diction. They share one structural feature. Each marks the point at which a limit exists in the larger description while the smaller one lacks the effective information to turn that limit into present, usable content. In one sentence: *limits determine what may become true; moduli and channels determine what can become usable*. The closure ladder studies the first, the rung geometry prices the traversal, and the population and strategic theory studies what happens when those prices and certificates are unevenly distributed among agents that interact.

14 LIMITATIONS AND OPEN ISSUES

The following are unresolved. They are stated as open, not as future work already under control.

1. **[Open] The motivating example is not covered.** The ladder begins at the computable sets and so presupposes a Turing-universal agent. The squirrel's gap is architectural, a gap of sub-universality, not of Turing degree. A resource-bounded analogue of the ladder is needed. The route we would try: replace $\leq_T$ by a sub-recursive reducibility and Turing ideals by ideals under it. The limit lemma has resource-bounded analogues, and the failure-of-idempotence argument behind Proposition 1 is indifferent to the reducibility, so a bounded ladder should exist; what is unclear is whether it stalls.
2. **[Open] Targets as decision problems.** Postulate 2 discards the perspectival axis entirely. Whether knowing *what it is like* admits a comparable formalisation, or is an independent dimension of cognitive space, is not addressed.
3. **[Open] Choice of approach.** Uniform, uncertified convergence is the load-bearing definitional choice. Non-uniform approach trivialises the closure (Lemma 2); certified approach freezes it (Proposition 1). Two alternatives are now excluded: bounded mind changes (Proposition 2) and probabilistic convergence (Proposition 3) generate the same ladder. Others, such as convergence with an oracle for a fixed non-arithmetical set, or non-uniform but resource-bounded approach, have not been compared.
4. **[Open] Minds as effective agents.** The results bind agents whose approximations are effectively generated. Physical hypercomputation, if realisable, moves every ceiling. Whether the physical Church–Turing thesis holds is not settled.
5. **[Open] Two axes.** Rungs here are measured by Turing degree. Proof-theoretic strength is a separate ordering, continuous in the sense of Definition 3 (Remark 4), with the same horizon as convergence by Feferman's completeness theorem for progressions (Section 9). A common refinement of the two orderings, if one exists, is not developed.
6. **[Open] Idealisation.** The results describe limits under unbounded resources. They characterise lineages of agents under scaling, not any single finite system.
7. **[Open] Connection and isolation between agents.** Relations between two agents' spaces (embedding, lateral separation via incomparable degrees, and causal, verificational and explanatory isolation) are defined informally only.
8. **[Open] No free rungs.**

Conjecture 1. *Every jump into an agent's void costs a downward channel: signal, verification or explanation.*

Neither a proof nor a counterexample is known to us.

9. **[Open] Self-verification of the next rung.** Theorem 3 shows that voids cannot be recognised by any observer, and that next-rung content can be verified in the limit from two rungs above. Whether an R_k -agent can verify membership of R_{k+1} in the limit from its own rung is open.
10. **[Open] An order parameter.** No measurable quantity distinguishing cognitive phases has been identified here; a candidate, the cluster order parameter of the accessibility-percolation conjecture, is proposed in [3].

11. **[Open] Quasi-isometry type of $\mathcal{A}$.** Whether $(\mathcal{A}, d)$ is quasi-isometric to a tree and whether its boundary at infinity is a single point are open problems of the companion paper [4], where tree-likeness is reduced to Manning’s bottleneck property for the graph of unit steps and refuted conditionally on an embedding hypothesis; that d is not itself a graph metric is settled there (Theorem 4(5)).
12. **[Open] Facts relied upon.** The randomness facts used in the companion paper are listed there. In this paper the cited facts, used in the relativised forms stated, are Gold’s marking theorem [14] in the form of Proposition 8, the theorem of Sacks and de Leeuw et al. in the relativised form used in Proposition 3, and Feferman’s completeness theorem for progressions in the form quoted in Section 9.
13. **[Open] Trust and testimony.** Certification from a higher rung reaches a lower agent only as testimony (Proposition 6(3)), which the recipient cannot verify even in the limit (Theorem 3(2)). The model says nothing yet about when accepting testimony is rational. Two routes are natural: a Bayesian treatment in which the prior over sources is itself limit-computable, and a reliabilist treatment in which testimony is accepted exactly when doing so does not increase the number of mind changes needed.
14. **[Open] Related work to engage.** Section 11 cites the neighbouring literature without yet arguing against it. The comparisons still to be made are: with Kelly’s limiting verification and refutation [19] and the later reliabilist work of Schulte and of Genin and Kelly, to state which of Theorem 3, Proposition 2 and Proposition 7 restate results there and which do not; with Weihrauch reducibility [6], where the closure ladder is the iteration of lim , to state what effective closure of a Turing ideal (the ideal structure, the agent reading, the stall) adds beyond $\text{lim}^{(n)}$; with Putnam and Gold read as epistemology rather than mathematics; with the philosophy of cognitive limits (McGinn [22], Chomsky’s distinction between problems and mysteries, Deutsch [10], and bounded rationality insofar as it touches item 1); and with the learning-theoretic literature on teaching behind Proposition 8.

15 CONCLUSION

This paper has fixed what a cognitive space is (a Turing ideal), what approach is (effective, uncertified convergence) and what follows. Iterated effective closure is the arithmetical ladder; the ladder exists because convergence is uncertified and freezes when a modulus is demanded; voids are generic; convergence stalls at ω ; a leap is undecidable from a process’s rules and unrecognisable from its behaviour by any observer, while next-rung content is recognisable in the limit from two rungs above; agents climb only by promotion; content crosses between agents as method, as testimony or as certification from above; and, in the one worked application, merger is the single certified route to new content inside a rung, and subsumption is limit-verifiable while strict superiority is not. The thread continues in three directions. The companion paper [4] takes the closure count as a distance and studies the geometry it induces on the arithmetical degrees, including the hidden-information theorem on which Section 13 draws. The sequel [2] puts the agents on a clock: populations, one-sided lag, finite computation under a cosmological constant, and the strategic question of what can be demonstrated, bluffed or audited ([2, Section 5]). The fourth paper [3] turns that dynamics into a quantitative model of the Fermi problem. A synopsis of the programme is [1].

POSTSCRIPT: ROOK’S TREE

An image, not an argument; it belongs to Figure 3 and to nothing proved above. *Picture a tree grown downwards from the sky (Figure 3). Its single trunk, $\emptyset, \emptyset', \emptyset'', \dots$, climbs for ever towards a crown it never reaches: ξ , true arithmetic, hung at the rim of the arithmetical world like a star at the end of a road. Every mind that convergence can build is a leaf somewhere on this tree. The generalised-low ones hang on its branches exactly. Each branch rises by the only means a mind has, taking in the canonical knowledge of the next rung, and at a height fixed by the mind’s own complexity it merges into the trunk and is lost in it. Two such minds are exactly as far apart as the knowledge needed to reconcile them, and their branches meet where that knowledge suffices. Every climb is a ray towards the crown, and every ray leans in towards the trunk; some touch it and vanish, and some climb beside it for ever, a single step away, never quite arriving. Yet the tree is not quite a tree. Give two distant minds the same education and they reach towards each other across the open air between the boughs, closer than any branch allows; the canopy fills with flats. And above the crown, beyond*

the last horosphere, the sky is empty. No amount of climbing reaches it. Only a jump crosses that silence, onto the next tree, which grows from the first tree's star.

AUTHOR CONTRIBUTIONS

The conception of the project, the framework of cognitive spaces, the motivating examples, the distinction between boundaries and voids, the analogy with undecidable spectral gaps, the conception of mastery as the capacity to teach and of the asymmetry of transmission across jumps, and the direction of the argument throughout are due to the author. The formalisation (the effective closure operator, the agent model, the statements and proofs of the lemmas, theorems and propositions, and the transfinite summary), the identification of the relevant literature and much of the drafting were developed with Claude, an AI system developed by Anthropic, used under the author's direction in extended dialogue. The encounter model of Section 8 (merger as elimination-and-replacement, the three outcomes) is due to the author; its formalisation and Theorems 5 and 6 were developed in dialogue. Several later additions (Corollary 1, Propositions 2 and 3, and the agent model as a pair of spaces) arose from a critical reading of an earlier version conducted by Claude at the author's request; the author has reviewed them. Claude is not listed as an author because it cannot take responsibility for the work, respond to review or correct the record. The author takes full responsibility for all claims, proofs and errors. The proofs were developed with an AI system and reviewed by the author; they have not been independently refereed.

REFERENCES

- [1] N. Goodman. Rungs, Voids and Effective Closure: a programme synopsis. Working paper (Paper O), 2026.
- [2] N. Goodman. Populations and strategy in cognitive spaces: lag, horizons, finite computation and the Gambler. Working paper (Paper C), 2026.
- [3] N. Goodman. Cognitive accessibility and the Fermi problem: a rate model, a Fermi function and a marked birth-and-contact process. Working paper (Paper D), 2026.
- [4] N. Goodman. A jump metric on the arithmetical degrees: rays, a shadow tree, dead ends, and the question of tree-likeness. Working paper (Paper A), 2026.
- [5] J. Bausch, T. S. Cubitt, A. Lucia, D. Pérez-García and M. M. Wolf. Size-driven quantum phase transitions. *Proc. Natl. Acad. Sci. USA* 115(1), 2018.
- [6] V. Brattka, G. Gherardi and A. Pauly. Weihrauch complexity in computable analysis. In *Handbook of Computability and Complexity in Analysis*, Springer, 2021.
- [7] G. J. Chaitin. Information-theoretic limitations of formal systems. *J. ACM* 21(3), 1974.
- [8] T. S. Cubitt, D. Pérez-García and M. M. Wolf. Undecidability of the spectral gap. *Nature* 528, 2015.
- [9] K. de Leeuw, E. F. Moore, C. E. Shannon and N. Shapiro. Computability by probabilistic machines. In *Automata Studies*, Princeton, 1956.
- [10] D. Deutsch. *The Beginning of Infinity*. Allen Lane, 2011.
- [11] Yu. L. Ershov. A hierarchy of sets I. *Algebra i Logika* 7, 1968.
- [12] S. Feferman. Transfinite recursive progressions of axiomatic theories. *J. Symbolic Logic* 27(3), 1962.
- [13] P. Gács. Every sequence is reducible to a random one. *Information and Control* 70, 1986.
- [14] E. M. Gold. Language identification in the limit. *Information and Control* 10, 1967.
- [15] E. M. Gold. Limiting recursion. *J. Symbolic Logic* 30(1), 1965.
- [16] M. H. Hart. An explanation for the absence of extraterrestrials on Earth. *Q. J. R. Astron. Soc.* 16, 1975.
- [17] S. M. Kautz. *Degrees of Random Sets*. PhD thesis, Cornell University, 1991.

- [18] A. Kučera. Measure, Π_1^0 -classes and complete extensions of PA. In *Recursion Theory Week*, Lecture Notes in Mathematics 1141, Springer, 1985.
- [19] K. T. Kelly. *The Logic of Reliable Inquiry*. Oxford University Press, 1996.
- [20] S. C. Kleene. Hierarchies of number-theoretic predicates. *Bull. Amer. Math. Soc.* 61, 1955.
- [21] C. Liu. *The Dark Forest*. Chongqing Press, 2008; English translation Tor, 2015.
- [22] C. McGinn. Can we solve the mind–body problem? *Mind* 98, 1989.
- [23] Y. N. Moschovakis. *Descriptive Set Theory*, 2nd edition. American Mathematical Society, 2009.
- [24] E. L. Post. Degrees of recursive unsolvability (abstract). *Bull. Amer. Math. Soc.* 54, 1948.
- [25] H. Putnam. Trial and error predicates and the solution to a problem of Mostowski. *J. Symbolic Logic* 30(1), 1965.
- [26] G. E. Sacks. On the degrees less than $\mathbf{0}'$. *Ann. of Math.* 77, 1963.
- [27] G. E. Sacks. A minimal degree less than $\mathbf{0}'$. *Bull. Amer. Math. Soc.* 67, 1961.
- [28] S. C. Kleene and E. L. Post. The upper semi-lattice of degrees of recursive unsolvability. *Ann. of Math.* 59, 1954.
- [29] G. E. Sacks. *Higher Recursion Theory*. Springer, 1990.
- [30] G. E. Sacks. *Degrees of Unsolvability*. Annals of Mathematics Studies 55, Princeton, 1963.
- [31] A. Sandberg, E. Drexler and T. Ord. Dissolving the Fermi paradox. arXiv:1806.02404, 2018.
- [32] J. R. Shoenfield. On degrees of unsolvability. *Ann. of Math.* 69, 1959.
- [33] R. I. Soare. *Turing Computability: Theory and Applications*. Springer, 2016.
- [34] C. Spector. Recursive well-orderings. *J. Symbolic Logic* 20, 1955.
- [35] A. Tarski. Der Wahrheitsbegriff in den formalisierten Sprachen. *Studia Philos.* 1, 1936.
- [36] A. M. Turing. Systems of logic based on ordinals. *Proc. London Math. Soc.* 45, 1939.

CONTACT

Nial Goodman • nialgoodman@gmail.com